Picking Up and Moving an Object

EGR 445/545: Project 2

Overview

For the second group project, your team will build and program a robot that can pick up an object from a certain (x, y, z) location and move it to a specified (x, y, z) destination while avoiding an obstacle.

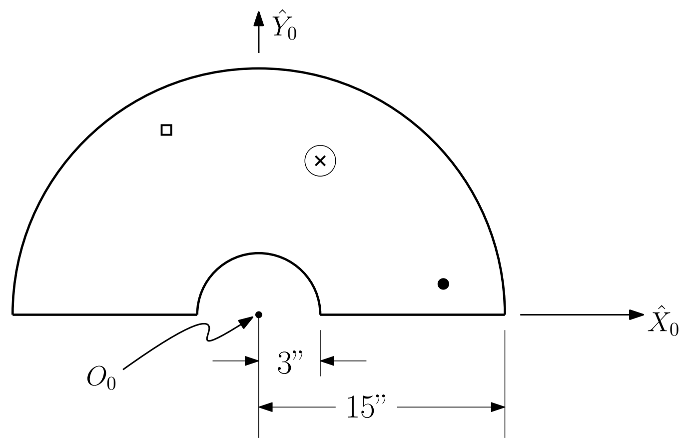


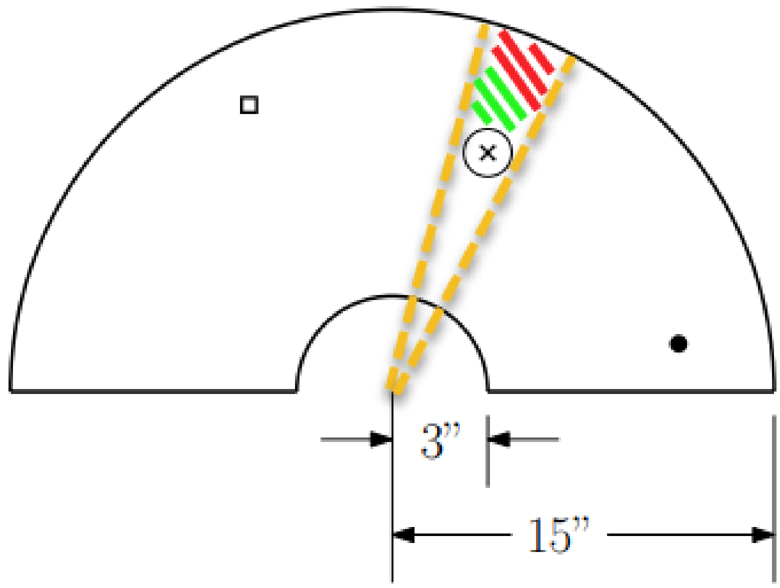
Figure 1: Schematic of the workspace, including the pick-up and drop-off locations and the obstacle.

Specifications

- Your code must take three inputs:
    1. the pick-up location
    2. the drop-off location
    3. the obstacle location
- Once the inputs are entered, the robot must perform the task in a fully automated fashion: no other inputs or human intervention are allowed.
- Your robot must be able to pick up and drop off the payload at any point in the workspace. The workspace is half of a donut shape (i.e. a cirle with a hole in the middle which is then cut in half). See Figure 1.
    - the inner radius is 3 in.
    - the outer radius is 15 in.
- The obstacle will have a maximum radius of 2 in.
- The payload will not exceed 5 ounces.

Rules Clarifications

- The five servos you have been issued are the only actuators allowed.
- You are not expected to reach around the obstacle tower. Neither the pick-up nor the drop-off point will be in the area that is radially behind the the obstacle:

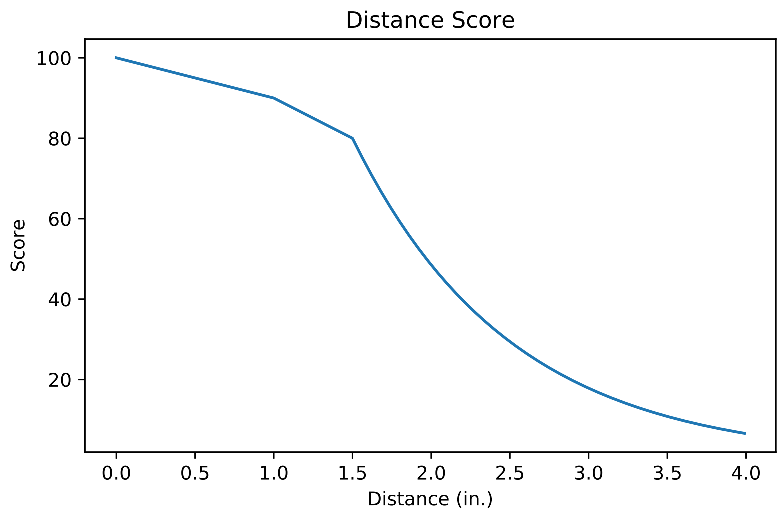

3"
15"

Grading

- inverse kinematics
- robot design
- end-effector design
- code
- performance
  - combination of accuracy and execution time

Performance Curves

Distance Score:



Time Score:

TBD

The new servos are slower and the time scores need to be determined.